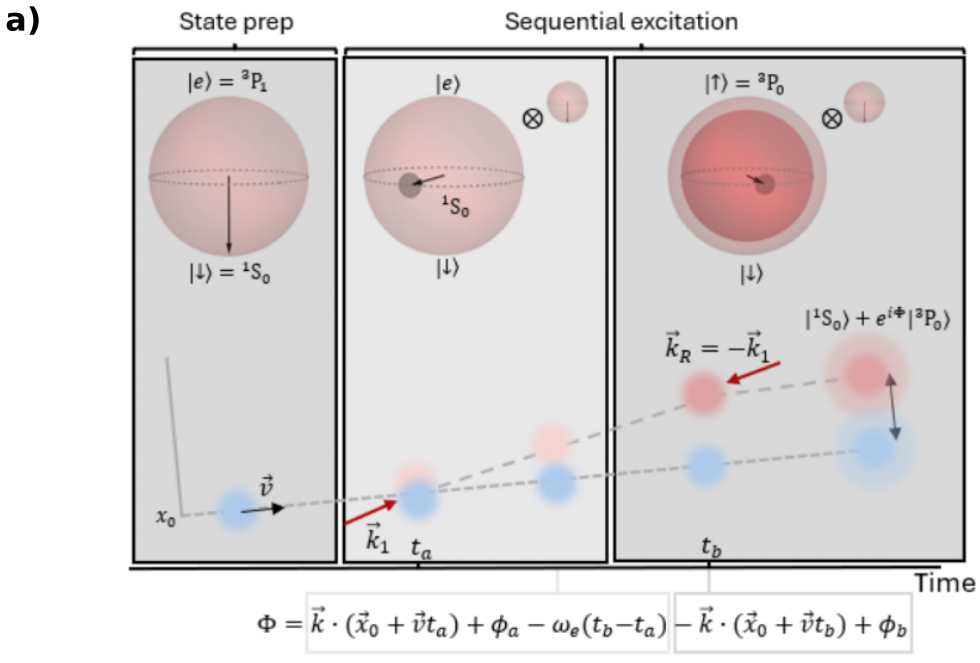
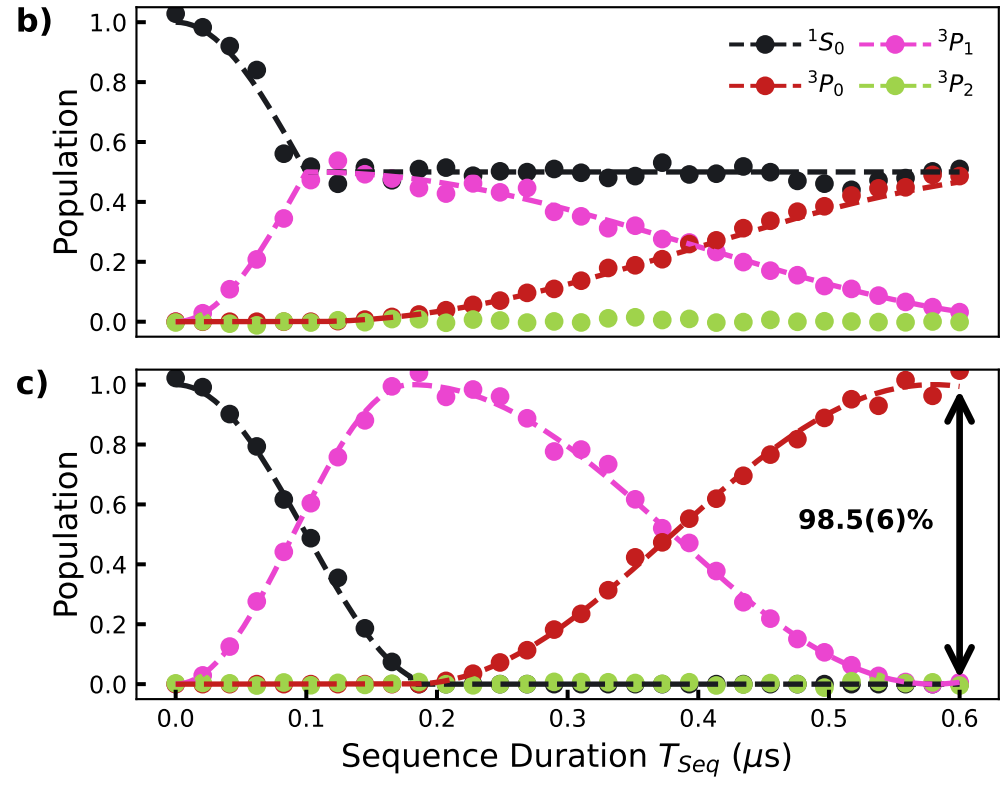

a)
State prep
Sequential excitation
$|e\rangle = {}^3P_1$
$|\downarrow\rangle = {}^1S_0$
$|e\rangle$
${}^1S_0$
$|\downarrow\rangle$
$|\uparrow\rangle = {}^3P_0$
$|\downarrow\rangle$
$|{}^1S_0\rangle + e^{i\Phi}|{}^3P_0\rangle$
$\vec{k}_R = -\vec{k}_1$
$\vec{v}$
x_0
$\vec{k}_1$
t_a
t_b
Time
$\Phi = \vec{k}\cdot(\vec{x}_0 + \vec{v}t_a) + \phi_a - \omega_e(t_b - t_a) - \vec{k}\cdot(\vec{x}_0 + \vec{v}t_b) + \phi_b$


b)
c)
Population
1.0
0.8
0.6
0.4
0.2
0.0
${}^1S_0$
${}^3P_1$
${}^3P_0$
${}^3P_2$
98.5(6)%
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Sequence Duration T_{Seq} (μs)